

=== GNB (1785496129) ===

```
# This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
# This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth
# To run the srsRAN Project gNB with this config, use the following command:
#  sudo ./gnb -c gnb_zmq.yaml
```

```
cu_cp:
  amf:
    addr: 10.53.1.2          # The address or hostname of the AMF.
    port: 38412
    bind_addr: 10.53.1.3    # A local IP that the gNB binds to for traffic from the AMF.
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "21401"
            tai_slice_support_list:
              - sst: 1
    inactivity_timer: 7200  # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported: [1 - 7200]

ru_sdr:
  device_driver: zmq        # The RF driver name.
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6 # Optionally pass argument
  srate: 11.52             # RF sample rate might need to be adjusted according to selected bandwidth.
  tx_gain: 75              # Transmit gain of the RF might need to adjusted to the given situation.
  rx_gain: 75              # Receive gain of the RF might need to adjusted to the given situation.

cell_cfg:
  dl_arfcn: 632000         # ARFCN of the downlink carrier (center frequency).
  band: 78                 # The NR band.
  channel_bandwidth_MHz: 20 # Bandwidth in MHz. Number of PRBs will be automatically derived.
  common_scs: 15           # Subcarrier spacing in kHz used for data.
  plmn: "21401"           # PLMN broadcasted by the gNB.
  tac: 7                   # Tracking area code (needs to match the core configuration).
  pdccch:
    common:
      ss0_index: 0          # Set search space zero index to match srsUE capabilities
      coresets0_index: 6    # Set search CORESET Zero index to match srsUE capabilities
    dedicated:
      ss2_type: common      # Search Space type, has to be set to common
      dci_format_0_1_and_1_1: false # Set correct DCI format (fallback)
  prach:
    prach_config_index: 1   # Sets PRACH config to match what is expected by srsUE
  pdsch:
    mcs_table: qam64        # Sets PDSCH MCS to 64 QAM
  pusch:
    mcs_table: qam64        # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log    # Path of the log file.
  all_level: info          # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false        # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap # Path where the MAC PCAP is stored.
  ngap_enable: false       # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap # Path where the NGAP PCAP is stored.
  e2ap_enable: true        # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

#e2:
#  enable_du_e2: true       # Enable DU E2 agent (one for each DU instance)
#  e2sm_kpm_enabled: true   # Enable KPM service module
```

```
# e2sm_rc_enabled: true          # Enable RC service module
# addr: 127.0.0.1                # RIC IP address
#bind_addr: 127.0.0.100         # A local IP that the E2 agent binds to for traffic from the RIC. ONLY required if
# port: 36421                    # RIC port

metrics:
  layers:
    enable_rlc: true            # Enable RLC metrics
    enable_sched: true          # Enable DU scheduler metrics
  periodicity:
    du_report_period: 1000      # Set DU statistics report period to 1s
```

=== UE (1785496129) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 23.04e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=23.04e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 077889900aabbccddeeff1122334455
imsi = 214010000000008

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1785496129) ===

```
#
# Copyright 2021-2026 Software Radio Systems Limited
#
# This file is part of srsRAN
#
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# it under the terms of the GNU Affero General Public License as
# published by the Free Software Foundation, either version 3 of
# the License, or (at your option) any later version.
#
# srsRAN is distributed in the hope that it will be useful,
# but WITHOUT ANY WARRANTY; without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the
# GNU Affero General Public License for more details.
#
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# the LICENSE file in the top-level directory of this distribution
# and at http://www.gnu.org/licenses/.
#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
    - SUBSCRIBER_DB=214010000000008,077889900aabbccddeeff1122334455,opc,63BFA50EE6523365FF14C1F45F88737D,8000,9,10.4
    - NETWORK_NAME_FULL=srsRAN
    - NETWORK_NAME_SHORT=srsRAN
    - TZ=Europe/Madrid
  privileged: true
  ports:
    - "9898:9999/tcp"
  # Uncomment port to use the 5gc from outside the docker network
  # - "38412:38412/sctp"
  # - "2152:2152/udp"
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.2}

gnb:
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
```

```

# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's the easiest
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
gnb_config.yml:
  content: |
    # This configuration file example shows how to configure the srsRAN Project gNB to allow srsUE to connect to it.
    # This specific example uses ZMQ in place of a USRP for the RF-frontend, and creates an FDD cell with 10 MHz bandwidth.
    # To run the srsRAN Project gNB with this config, use the following command:
    #   sudo ./gnb -c gnb_zmq.yml

  cu_cp:
    amf:
      addr: 10.53.1.2          # The address or hostname of the AMF.
      port: 38412
      bind_addr: 10.53.1.3    # A local IP that the gNB binds to for traffic from the AMF.
      supported_tracking_areas:
        - tac: 7
          plmn_list:
            - plmn: "21401"
              tai_slice_support_list:
                - sst: 1
      inactivity_timer: 7200  # Sets the UE/PDU Session/DRB inactivity timer to 7200 seconds. Supported values are 7200, 1800, 900, 450, 225, 112.5, 56.25, 28.125, 14.0625, 7.03125, 3.515625, 1.7578125, 0.87890625, 0.439453125, 0.2197265625, 0.10986328125, 0.054931640625, 0.0274658203125, 0.01373291015625, 0.006866455078125, 0.0034332275390625, 0.00171661376953125, 0.000858306884765625, 0.0004291534423828125, 0.00021457672119140625, 0.00010728836059570312, 0.00005364418029785156, 0.00002682209014892578, 0.00001341104507446289, 0.000006705522537231445, 0.000003352761268615722, 0.000001676380634307861, 0.0000008381903171539306, 0.0000004190951585769653, 0.0000002095475792884826, 0.0000001047737896442413, 0.00000005238689482212065, 0.00000002619344741106032, 0.00000001309672370553016, 0.00000000654836185276508, 0.00000000327418092638254, 0.00000000163709046319127, 0.000000000818545231595635, 0.0000000004092726157978175, 0.0000000002046363078989087, 0.0000000001023181539494544, 0.0000000000511590769747272, 0.0000000000255795384873636, 0.0000000000127897692436818, 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```

```

tac: 7                                # Tracking area code (needs to match the core configuration).
pdcch:
  common:
    ss0_index: 0                      # Set search space zero index to match srsUE capabilities
    coresets0_index: 6                # Set search CORESET Zero index to match srsUE capabilities
  dedicated:
    ss2_type: common                  # Search Space type, has to be set to common
    dci_format_0_1_and_1_1: false    # Set correct DCI format (fallback)
prach:
  prach_config_index: 1               # Sets PRACH config to match what is expected by srsUE
pdsch:
  mcs_table: qam64                   # Sets PDSCH MCS to 64 QAM
pusch:
  mcs_table: qam64                   # Sets PUSCH MCS to 64 QAM

log:
  filename: /tmp/gnb.log              # Path of the log file.
  all_level: info                     # Logging level applied to all layers.
  hex_max_size: 0

pcap:
  mac_enable: false                  # Set to true to enable MAC-layer PCAPs.
  mac_filename: /tmp/gnb_mac.pcap    # Path where the MAC PCAP is stored.
  ngap_enable: false                 # Set to true to enable NGAP PCAPs.
  ngap_filename: /tmp/gnb_ngap.pcap  # Path where the NGAP PCAP is stored.
  e2ap_enable: true                  # Set to true to enable E2AP PCAPs.
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap # Path where the DU E2AP PCAP is stored.
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap # Path where the CU-CP E2AP PCAP is stored.
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap # Path where the CU-UP E2AP PCAP is stored.

#e2:
#  enable_du_e2: true                # Enable DU E2 agent (one for each DU instance)
#  e2sm_kpm_enabled: true             # Enable KPM service module
#  e2sm_rc_enabled: true              # Enable RC service module
#  addr: 127.0.0.1                   # RIC IP address
#  #bind_addr: 127.0.0.100           # A local IP that the E2 agent binds to for traffic from the RIC. ONLY required for E2SM
#  port: 36421                       # RIC port

metrics:
  layers:
    enable_rlc: true                 # Enable RLC metrics
    enable_sched: true               # Enable DU scheduler metrics
  periodicity:
    du_report_period: 1000           # Set DU statistics report period to 1s
gnb_compose_config.yml:
content: |
  cu_cp:
    amf:
      addr: ${OPEN5GS_IP:-10.53.1.2}
      bind_addr: 0.0.0.0
  metrics:
    autostart_stdout_metrics: true
    enable_json: true
  remote_control:
    bind_addr: 0.0.0.0
    enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:

```

driver: default

config:

- subnet: 172.19.1.0/24

=== NOTAS (1785496129) ===

1. PLMN (21401) sincronizado en gNB, UE y Core. (REGLA 1)
2. dl_arfcn (632000) seleccionado para Banda 78 y 20 MHz. (REGLA 2)
3. bind_addr del gNB en docker-compose ajustado a 0.0.0.0. (REGLA 5)
4. Bloque ru_sdr del gNB copiado exactamente del CAG. (REGLA 6)
5. IMSI, K y OPC sincronizados en UE y SUBSCRIBER_DB del Core. (REGLA 7)
6. K del suscriptor corregido a 32 caracteres hexadecimales (077889900aabbccddeeff1122334455). (REGLA 7a)
7. srate y base_srate del UE ajustados a 23.04e6 para 20 MHz de ancho de banda.